THE JOURNAL OF PHYSICAL CHEMISTRY B
A JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
pubs.acs.org/JPCB
Article

Dynamics of $Emim^+$ in [Emim][TFSI]/LiTFSI Solutions as Bulk and under Confinement in a Quasi-liquid Solid Electrolyte

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Cite This: J. Phys. Chem. B 2021, 125, 5443−5450
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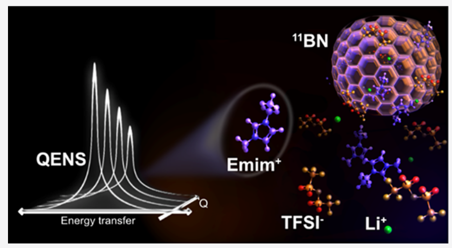
ABSTRACT: Quasi-liquid solid electrolytes are a promising alternative for next-generation Li batteries. These systems combine the safety of solid electrolytes with the desired properties of liquids and are typically formed by solutions of Li salts in ionic liquids incorporated into solid matrices. Here, we present a fundamental understanding of the transport properties in solutions of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) in 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([Emim][TFSI]), either in bulk form or incorporated in a boron nitride (BN) matrix. We performed a series of quasi-elastic neutron scattering experiments that, given the high incoherent neutron scattering cross section of hydrogen, allowed us to focus on the $Emim^+$ dynamics. First, [Emim][TFSI]/LiTFSI solutions (0.5 and 2.5 $mol \cdot kg^{-1}$) were investigated and we show how the increase in the concentration reduces the $Emim^+$ mobility and increases the activation energy of their long-range motions. Then, the 0.5 $mol \cdot kg^{-1}$ solution was incorporated into the BN matrix and we report that the diffusivities of the $Emim^+$ cations that remain mobile under confinement are highly accelerated in comparison with the bulk sample and the activation energy of these motions is drastically reduced. We present the experimental evidence that this effect is related to the content of the $Emim^+$ cations immobilized near the surfaces of the BN pores.



■ INTRODUCTION

The conception and commercialization of lithium-ion batteries (LIBs) in the 1990s have played a major role in shaping our modern society. LIBs are among the technologies responsible for bringing several portable devices into our daily lives, and their path toward larger-scale applications is being rapidly expanded.[1,2] Nevertheless, the continuous increase in the use of these batteries raises a series of debates related to the limits imposed by the properties of the current commercially available devices.[3] Among several concerns, frequent discussions occur related to the safety of traditional liquid electrolytes formed by solutions of $LiPF_6$ in organic solvents (mostly cyclic carbonates).[4] While this formulation possesses an exquisite film-forming ability and, consequently, prevents rapid degradation of the aluminum current collectors and enables reversible cycling, they are also highly flammable, toxic, and chemically unstable.[5]

Currently, the substitution of organic solvents with ionic liquids (IL) is considered one of the most promising alternatives to mitigate the above-listed limitations.[6] The so-called molten salts exhibit nonflammability, negligible vapor pressure, and high chemical and thermal stabilities, which are combined with the ability to dissolve high contents of Li salts.[7] Nevertheless, even with these remarkable features and several encouraging results obtained during the past decade with battery prototypes,[8] the performance of the IL/Li salt electrolytes alone is limited by their high viscosities and consequent low ionic conductivities.[9] Solid electrolytes are also considered an attractive alternative to the current state-of-the-art liquid electrolytes as they can substantially mitigate risks related to leakage, volatilization, and flammability.[10] However, the use of these systems is often limited by their poor ionic conductivity and lithium dendrite formation.[11,12]

Ultimately, the incorporation of IL/Li salt solutions in solid electrolytes has proven to be effective in improving their ionic conductivity and reducing the resistance at the electrode/electrolyte interfaces. In polymer-based solid electrolytes, however, these benefits are compromised by poor performance at room temperature due to discontinuous transport of Li ions as caused by the limited uptake of the IL/Li electrolytes.[13] Facing these limitations, inorganic particles/matrices have been used in combination with IL/Li electrolytes as a strategy to provide efficient pathways for the transport of lithium ions.[14−16]

Among the recently investigated inorganic matrices, 2D graphene-analogue boron nitride (BN) provides a set of unique features for solid electrolyte applications.[17] In addition to being electronically insulating, the synthesis can be tailored to produce a high surface area, porosity, and mechanical modulus that allows for the accommodation of large volumes

Received: March 16, 2021
Revised: May 2, 2021
Published: May 18, 2021

https://doi.org/10.1021/acs.jpcb.1c02383
J. Phys. Chem. B 2021, 125, 5443−5450

of IL/Li liquid solutions while the solid-state character is preserved. Moreover, the layered configuration of BN provides favorable paths for ionic migration leading to satisfactory ionic conductivity. With these concepts, one can define the combination between the solid BN matrix and the IL/Li liquid solutions as a quasi-liquid solid electrolyte (QLSE), where the resulting material offers the safety of a solid-state electrolyte together with advantages that are inherent from liquid systems.[17]

In recent works, the applicability of the QLSE in batteries has been successfully demonstrated.[17,18] Yet, there has been little fundamental effort to understand the transport properties and the nature of conduction in these electrolytes. In the present work, we have combined the ionic liquid 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide ([Emim][TFSI]) with lithium bis(trifluoromethanesulfonyl)-imide (LiTFSI) to compose the liquid portion of a BN-based QLSE. With this combination, as an aprotic IL is used, the $TFSI^-$ anions may preferably engage in the coordination of Li^+ rather than with the $[Emim]^+$ cations, which may bring direct outcomes to the dynamical behavior of the ionic species within the system and at last to the QLSE's properties. Also, the interactions between the liquid solution and the solid surfaces may be complex and directly influence the performance of QLSE-containing energy devices. Hence, tuning the QLSEs toward their optimum efficiency is only possible with a rational understanding of the transport properties of each of the mobile components within the electrolytes.

Here, we focus on the description of the transport properties of the $[Emim]^+$ cations, which, as hydrogen-rich molecules, can be readily investigated with quasi-elastic neutron scattering (QENS), as demonstrated in previous works.[19−21] With QENS, one can benefit not only from the high neutron incoherent cross section of ^1H atoms but also from the high penetration of neutrons into matter.[22] Due to the latter, H-rich fluids and molecules under confinement in very diverse matrices have been extensively investigated with QENS.[23−26] To explore this feature, due to the high neutron absorption cross section of ^{10}B, the BN matrix was prepared with ^{11}B enrichment to form ^{11}BN.

First, by the analysis of bulk solutions of LiTFSI in [Emim][TFSI], we have been able to describe the outcomes in the dynamics of the $Emim^+$ cations due to a large increase in the molality, from 0.5 to 2.5 $mol{\cdot}kg^{-1}$. Then, the solution with a concentration of 0.5 $mol{\cdot}kg^{-1}$ was confined within the ^{11}BN matrix, and the implications to the $Emim^+$ dynamics have been described. We report that, under confinement, these cations perform faster long-range motions with a considerable reduction in the activation energy as compared with the bulk liquid. Our data suggest that this effect is related to the formation of a protective layer of immobilized species near the inner pore surfaces as well as to the specific presence of the Emim molecules in this process.

■ METHODS

Sample Preparation. For the fabrication of the ^{11}BN matrix, we followed the method described in ref 17. Boron oxide (B_2O_3) and urea (CON_2H_4) were dissolved in a mixture of ethanol and ultrapure water (67 vol % of ethanol) and stirred at 70 °C until recrystallization. For ^{11}BN, $^{11}B_2O_3$ was used as the starting material. The resulting solid was gradually heated in a tube furnace to 900 °C with a 4 $°C{\cdot}min^{-1}$ heating rate and under a N_2 atmosphere. The furnace was further kept at 900 °C for 2 h and subsequently cooled to room temperature, leading to the formation of nanoporous ^{11}BN powder.

For the preparation of the IL, [EMIM/TFSI], the route described in ref 27 has been followed. Then, LiTFSI powder was added to the IL to achieve solutions of molalities 0.5 and 2.5 $mol{\cdot}kg^{-1}$. Hereafter, these solutions will be named as [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI-[2.5m]. [Emim][TFSI]/LiTFSI[0.5m] was used as the liquid portion of the QLSE and was mixed with the ^{11}BN matrix. In a prior work,[17] a weight ratio of 1:10 of BN:IL was used, whereas for this experiment, a ratio of approximately 1:1 was used to ensure that all of the IL was confined within the ^{11}BN matrix. Homogeneous mixing of the IL with ^{11}BN was achieved by dissolving the IL in anhydrous ethanol followed by agitation and then removal of the ethanol in a vacuum oven. The sample was heated to 80 °C to complete the confinement of the IL.

Adsorption Isotherms. A TriStar 3000 volumetric adsorption analyzer (Micromeritics Instrument Corp., USA) was used to measure the nitrogen adsorption isotherms at 77 K in the ^{11}BN matrix as well as in a sample of BN for comparison purposes. Before the experiments, the samples were dried under nitrogen gas flow.

Quasi-elastic Neutron Scattering (QENS) Experiments. QENS experiments were performed on the back-scattering spectrometer BASIS at the Spallation Neutron Source, Oak Ridge National Laboratory, USA.[28] With the BASIS, an energy resolution of ∼3.7 μeV (full width at half-maximum for the Q-averaged resolution value) can be achieved with a dynamic range of accessible energy transfer values of ±100 μeV. For the experiments, the samples were placed in flat aluminum plate sample holders and positioned perpendicularly to the neutron beam in the spectrometer. Full QENS spectra were collected at 10, 290, 310, 330, 350, and 370 K for the [Emim][TFSI]/LiTFSI[0.5m] and ^{11}BN-containing [Emim]-[TFSI]/LiTFSI[0.5m] samples. The data collected at 10 K were used as the instrument resolutions in the data analysis. Data from dry ^{11}BN were also collected at 10, 290, 310, 330, 350, and 370 K, and the obtained results were subtracted from the ^{11}BN-containing [Emim][TFSI]/LiTFSI[0.5m] for the exclusion of the background signal. In the case of the [Emim][TFSI]/LiTFSI[0.5m] data, the exclusion of the background signal was performed by subtracting data collected at 370 K with an empty aluminum plate. QENS spectra from the [Emim][TFSI]/LiTFSI[2.5m] solution were collected at 10, 340, 355, and 370 K. For this sample, the exclusion of the background signal was also performed by subtracting the data from the empty aluminum plate. The ^{11}BN-containing [Emim][TFSI]/LiTFSI[0.5m] was also subjected to temperature-dependent elastic intensity scans between 20 and 320 K with 1 K steps upon heating and cooling while the bulk electrolyte was subjected to a similar analysis upon heating.

QENS Data Analysis. The collected data were reduced using Mantid software[29] and further analyzed using DAVE software.[30] In general, the measured intensities in QENS experiments, $I(Q, E)$, are well described by

$$I(Q, E) = [A_0(Q)\delta(E) + (1 - A_0(Q))S(Q, E)] \otimes R(Q, E) + B(Q, E) \quad (1)$$

where $A_0(Q)$ is the elastic fraction of the scattering signal, $\delta(E)$ is a Dirac delta function that accounts for the elastic scattering (zero energy transfer), $S(Q, E)$ is the dynamic structure factor

https://doi.org/10.1021/acs.jpcb.1c02383
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(or dynamic scattering function), $R(Q, E)$ corresponds to the instrument resolution function, and $B(Q, E)$ is a linear background. $S(Q, E)$ has been described by a Cole–Cole distribution function:[31]

$$S(Q, E) = \frac{1}{\pi\Gamma(Q)} \frac{\left(\frac{E-E_0}{\Gamma(Q)}\right)^{-\alpha}\cos\left(\frac{\pi\alpha}{2}\right)}{1 + 2\left(\frac{E-E_0}{\Gamma(Q)}\right)^{1-\alpha}\sin\left(\frac{\pi\alpha}{2}\right) + \left(\frac{E-E_0}{\Gamma(Q)}\right)^{2(1-\alpha)}} \quad (2)$$

In eq 2, $\Gamma(Q)$ is the half width at half-maximum of the curve, and α is a stretching factor ($0 \leq \alpha \leq 1$). In the lower limit case, $\alpha = 0$, eq 2 becomes a Lorentzian function. The stretching factor accounts for the presence of a distribution of the dynamic motions around the mean value. For the bulk ionic liquids, no contribution of elastic scattering in the QENS data has been considered ($A_0(Q) = 0$ in eq 1). For the ^{11}BN-containing [Emim][TFSI]/LiTFSI[0.5m], $A_0(Q)$ was allowed to vary to account for the contributions from the $Emim^+$ ions immobilized or drastically slowed down (below the energy resolution of the spectrometer) due to the interactions with the solid matrix.

The broadenings of the QENS signals, $\Gamma(Q)$, were further analyzed for their Q dependence. For the bulk samples, the data were well described by the $\hbar DQ^2$ law of macroscopic dynamics, with D being the diffusion coefficient. For the ^{11}BN-containing [Emim][TFSI]/LiTFSI[0.5m], the $\Gamma(Q)$ was leveling off, instead of approaching zero, in the limit of low Q, and hence the fitting of the $\Gamma(Q)$ was performed by considering the effect of confinement using the expression[32]

$$\Gamma(Q) = \left(\frac{4.33\hbar D}{a^2}\right)\left[1 - \frac{1}{1 + \exp\left(-H\left(Q - \frac{3.3}{a}\right)\right)}\right] + \left(\frac{\hbar DQ^2}{1 + DQ^2\tau_0}\right)\left[1 - \frac{1}{1 + \exp\left(-H\left(Q - \frac{3.3}{a}\right)\right)}\right] \quad (3)$$

Here, τ is the residence time between jumps amid two sites, D is the diffusion coefficient, a is the confinement radius experienced by the mobile species, and the terms in square brackets are modulation terms defining the region where the QENS signal is affected by the confinement (at low Q, such that $Q < 3.3/a$).

Finally, the temperature dependence of the diffusion coefficients obtained by fitting $\Gamma(Q)$ was evaluated using the Arrhenius equation

$$D = D_0 e^{-E_{act}/(RT)} \quad (4)$$

with D_0 being a prefactor, R being the gas constant, and T being the absolute temperature.

RESULTS AND DISCUSSION

Before carrying out the confinement of the ionic liquid for the QENS experiments, physisorption measurements with N_2 isotherms were carried out to examine the presence of pores and estimate the surface area. The results of the analysis of the pore size distributions from the desorption data by the Barrett–Joyner–Halenda method are shown in Figure 1, and the surface areas from the Brunauer–Emmett–Teller (BET) method are shown in the inset. The BN and ^{11}BN samples display nearly identical pore size distributions with a population of pores with diameters ∼4 nm and broad distribution of pores with diameters of >10 nm. The surface areas for the BN and ^{11}BN samples are also identical and have been determined as 86 and 89 m^2/g, respectively. The ^{11}BN sample was used to confine the [Emim][TFSI]/LiTFSI[0.5m] solution.

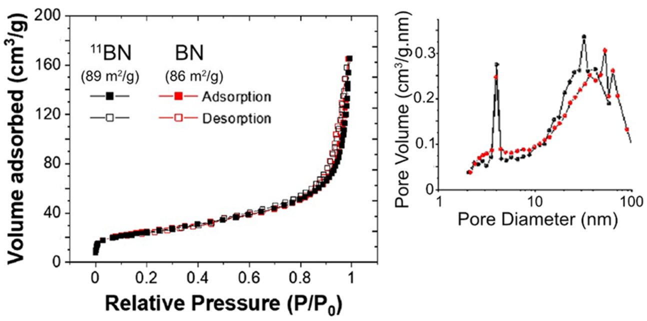

Figure 1. N_2 sorption isotherms and extracted pore size distributions of BN and ^{11}BN, which exhibited similar surface areas and pore size distributions.

For the bulk samples, [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m], a comparison between the normalized QENS signals at 370 K and $Q = 0.7$ Å^{-1} is presented in Figure 2a. Here, the reduction in the QENS broadening allows for appreciating the slowdown in the $Emim^+$'s dynamics as caused by the drastic increase in LiTFSI concentration and the likely more frequent coordination of $Emim^+$ by $TFSI^-$. The QENS data of all samples and the corresponding fits with eqs 1 and 2 are presented in the Supporting Information (Figures S1–S3).

From the analysis of the dynamic scattering functions with eq 2, the stretching factor α has been determined and is presented in Figure 2b for the data collected at 370 K. For both [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m], α is fairly Q-independent, presents an average value of $\overline{\alpha} = 0.24$ for the first sample, and increases to $\overline{\alpha} = 0.33$ for the more concentrated sample. This result goes along with the slowdown in the dynamics displayed in Figure 2a as the more frequent coordination of $Emim^+$ by $TFSI^-$ leads to larger fluctuations in the dominant signal. Recently, González et al. reported similar outcomes, regarding both the diffusion coefficients and the stretching of the QENS signal, in a work combining QENS and molecular dynamics simulations to characterize the dynamic properties of water in a water-in-salt electrolyte with different concentrations.[33]

Also, by fitting the QENS signals with eqs 1 and 2, the broadening of the curves $\Gamma(Q)$ has been obtained, and their Q^2 dependence has been evaluated as presented in Figure 2c,d. In the figures, the dotted lines represent the fittings with the macroscopic diffusion equation ($\Gamma(Q) = \hbar DQ^2$) and the so-determined diffusion coefficients, D, are displayed in Table 1 for the different temperatures in the experiment. While the temperature dependence of D will be discussed further in the manuscript together with the values obtained for the confined ionic liquids, it is noticeable from the figures that the $\Gamma(Q)$ values are severely damped, and so are the diffusion coefficients, due to the increase in LiTFSI concentration. As

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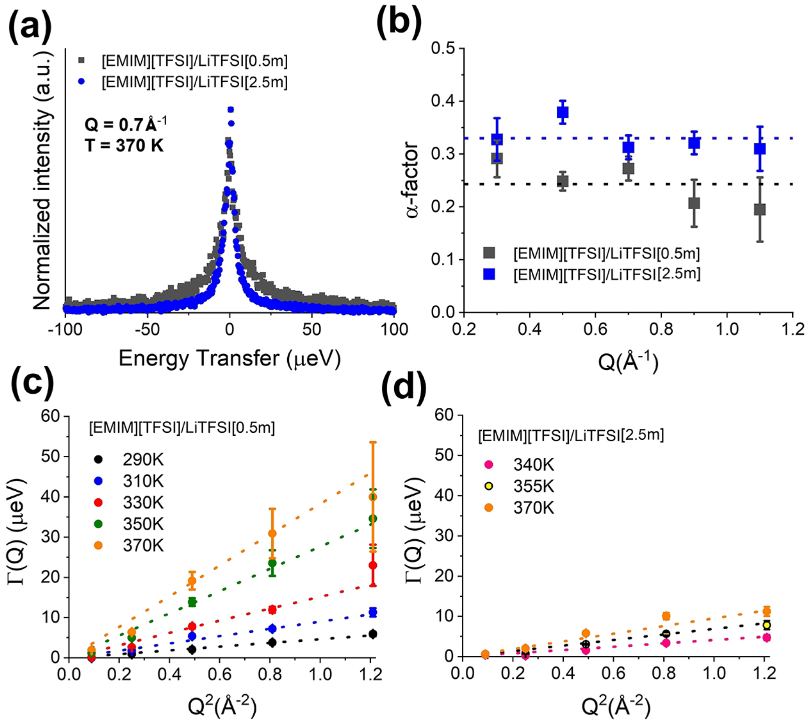

Figure 2. (a) Normalized QENS signals at 370 K and $Q = 0.7$ Å^{-1} for the [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m] bulk samples. (b) Q dependence of the stretching factor α at 370 K for the [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m] samples as determined by fitting the dynamic scattering functions with eq 2. (c, d) Q^2 dependence of the broadening of the dynamic scattering functions, $\Gamma(Q)$, for the [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m] samples at different temperatures.

Table 1. Parameters Obtained from the Fits of the QENS Broadenings of [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m] Bulk Samples with eqs 1 and 2

[Emim][TFSI]/LiTFSI[0.5m]		[Emim][TFSI]/LiTFSI[2.5m]	
temperature (K)	D (10^{-10} m^2/s)	temperature (K)	D (10^{-10} m^2/s)
290	0.71 ± 0.04	340	0.62 ± 0.03
310	1.36 ± 0.06	355	1.05 ± 0.06
330	2.31 ± 0.09	370	1.44 ± 0.28
350	4.20 ± 0.24		
370	5.82 ± 0.55		

one can observe in the data presented in Table 1, while the LiTFSI concentration has been increased by a factor of 5 (from 0.5 to 2.5 mol·kg^{-1}), the values of D are reduced by a factor of ~4 regardless of the temperature.

Turning to the analysis of the confined ionic liquid, that is, the ^{11}BN-containing [Emim][TFSI]/LiTFSI[0.5m], we initially present the results obtained with the temperature-dependent elastic scans. In Figure 3a, where the scans obtained upon cooling and heating are presented as sums of the data collected between $Q = 0.5$ and 1.1 Å^{-1}, there is no indication of step-like first-order transitions and, consequently, of the species with which bulk-like dynamics could be associated. Therefore, the Emim$^+$ cations under analysis are either immobilized (or slowed down below the limit of the instrument resolution) due to strong interactions with the ^{11}BN surfaces or subjected to confined geometries that preclude first-order phase transitions. While the former cations contribute to the elastic fraction of the signal, the latter are responsible for the dynamic signal measured in the experiment. The elastic scans of the confined electrolyte can be compared with the elastic scans collected with the bulk electrolyte upon heating, as presented in the inset of Figure 3a. For the bulk electrolyte, there is also no clear evidence of first-order phase transitions, indicating that the Emim$^+$ cations do not move strictly as free molecules in the media regardless of the confined geometry imposed by BN. For the bulk, around 70% of the Emim$^+$ cations contribute to motions within the instrumental resolution, as depicted by the drop in elastic intensity, while for the confined electrolyte, this percentage decreases to around 40%. At about 207 K, the measurable activation of motions is detected in the confined electrolyte, as exposed by the break in the monotonical decrease in the elastic intensities, and the superposition of the data collected upon cooling and heating shows that these events are reversible despite the confined geometry. For the bulk electrolyte, the motions are activated at higher temperatures, ~240 K, as a consequence of the less constrained geometry.[34] Considering the BET results presented in Figure 1 together with the elastic scans, it is unlikely that the dynamic scattering originates from motions of free molecules within the pores with diameters of >10 nm, as these would be even more equivalent to the dynamics of the bulk sample,[35,36] although the presence of the elastically scattering surface-immobilized ions in the >10 nm pores is possible. Also, considering the interlayer spacing of BN, that is, 3−4.5 Å,[17] one can assume that any dynamics from occasionally intercalated Emim$^+$ cations would not be fast enough to contribute to the QENS signal beyond the elastic fraction.[35,36]

Then, full QENS spectra were collected with the ^{11}BN-containing [Emim][TFSI]/LiTFSI[0.5m] at selected temperatures and fitted with eqs 1 and 2. In this case, however, $A_0(Q)$ was allowed to vary in eq 1 (as previously mentioned), and the stretching factor α converged to zero in eq 2, leading to a

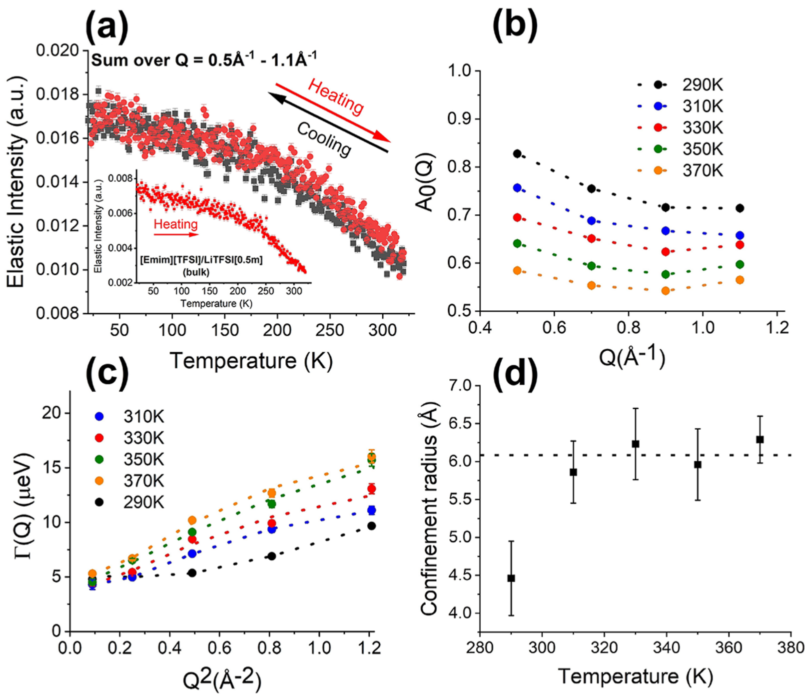

Figure 3. QENS results for the [11]BN-containing [Emim][TFSI]/LiTFSI[0.5m]. (a) The temperature-dependent elastic scans upon cooling and heating (sum over $Q = 0.5{-}1.1$ Å^{-1}) are presented, and the inset shows the elastic scan collected with the bulk electrolyte upon heating. (b) The Q dependences of $A_0(Q)$ at the different experimental temperatures are shown. (c) The Q^2 dependences of $\Gamma(Q)$ at different temperatures are displayed. In (b), the dotted lines are guides for the eyes, and in (c), the dotted lines show the fits of the data with eq 3. (d) The confinement radii are presented as a function of temperature. The dotted line shows the average value for the radii at temperatures above 290 K.

Table 2. Parameters Obtained from the Fits of QENS Broadenings of [11]BN-Containing [Emim][TFSI]/LiTFSI[0.5m] Obtained with eq 3

temperature (K)	confinement radius (Å)	D (10^{-10} m^2/s)	τ (ps)	H
290	4.46 ± 0.49	4.01 ± 0.70	45.63 ± 6.07	3.12 ± 1.46
310	5.86 ± 0.41	5.47 ± 0.49	43.34 ± 3.35	6.48 ± 2.98
330	6.23 ± 0.47	6.46 ± 0.60	37.54 ± 4.70	4.48 ± 2.06
350	5.96 ± 0.47	7.05 ± 0.56	30.90 ± 2.27	4.5[a]
370	6.29 ± 0.31	7.87 ± 0.54	31.02 ± 2.71	5.64 ± 1.84

[a]The value was fixed during the fitting due to divergence.

limiting case of a Lorentzian function. Note that α reflects the complexity of the chemical environment in which the Emim$^+$ cations are inserted and is, therefore, dependent on the frequency that Emim−Emim interactions as well as interactions between Emim$^+$ and other chemical species occur within the electrolyte. After the confinement of the electrolyte, as part of the Emim$^+$ cations are immobilized by the matrix walls, the complexity of the relaxations that remain detectable is expected to diminish, leaning α toward zero. The QENS data obtained with the [11]BN matrix and used for background subtraction are displayed in Figure S4 (Supporting Information). In Figure 3b, the Q dependence of $A_0(Q)$ is presented for the different experimental temperatures. Here, as the temperature is raised, a decrease in $A_0(Q)$ occurs, from 0.71 at 290 K to 0.56 at 370 K (considering the values at $Q = 1.1$ Å^{-1}). Therefore, we may infer that more than half of the Emim$^+$ ions are very strongly immobilized due to the confinement and do not perform detectable motions even at 370 K. Meanwhile, the drop in $A_0(Q)$ from 290 to 370 K indicates that an additional population of approximately one-fifth of the Emim$^+$ ions is subjected to strong interactions within the confined geometry but, upon the increase in temperature, can perform motions detectable beyond the instrumental resolution.

The Q^2 dependences of $\Gamma(Q)$ are presented in Figure 3c for the [11]BN-containing [Emim][TFSI]/LiTFSI[0.5m] at different temperatures, and Table 2 summarizes the parameters obtained by fitting the data with eq 3 (in the figure, the dotted lines represent the fits). Different from the bulk samples, the $\Gamma(Q)$ presents nonzero intercepts, which is consistent with the dynamics of liquids in constrained geometries.[32,37] As displayed in Figure 3d and Table 2, the confinement radius at 290 K has been determined as 4.46 ± 0.49 Å and is temperature-independent for the higher temperatures with an average value of 6.08 ± 0.84 Å. Considering the BET analysis (Figure 1), where a population of pores with diameters around 4 nm has been detected, the smaller but still qualitatively comparable dimensions of confinement as determined via QENS likely reflect the decoration of the inner surfaces of the pores by the components of the ionic liquids. Nevertheless, as the elastic fraction of Emim$^+$, $A_0(Q)$, decreases with temperature and no changes in the confinement radius are observed, other species (most likely TFSI$^-$) possibly take place in the decoration of the pores. The consequences of these effects on the dynamics of the confined fluid are discussed in the sequence.

In Figure 4a, the Arrhenius plot depicts the temperature dependence of the diffusion coefficients of the bulk samples as well as the confined ionic liquid. The respective activation energies are also presented, as determined via eq 4. As shown in the comparison between the [Emim][TFSI]/LiTFSI[0.5m] and [Emim][TFSI]/LiTFSI[2.5m] bulk samples, the drastic increase in LiTFSI concentration brings, as a consequence, a substantial reduction in the overall diffusivities of the $Emim^+$ cations. Additionally, an increase in the energy necessary to activate these motions is observed.

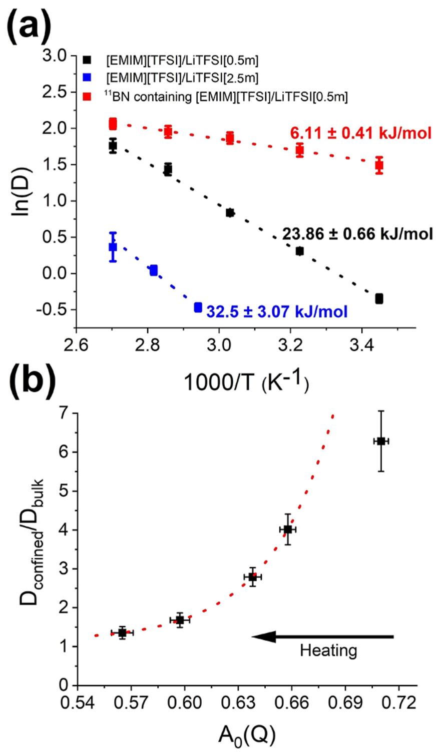

Figure 4. (a) An Arrhenius plot depicting the temperature-dependence of the diffusion coefficients of the bulk samples as well as the confined ionic liquids is presented. The respective activation energies are also shown in the figure. (b) We present the evolution of $D_{\mathrm{confined}}/D_{\mathrm{bulk}}$ for [Emim][TFSI]/LiTFSI[0.5m] as a function of the elastic fraction of $Emim^+$ ($A_0(Q)$) as determined for the confined fluid. The red dotted lines are exponential growth curves.

As we turn to the comparison between the diffusion coefficients from the bulk [Emim][TFSI]/LiTFSI[0.5m] and those from the confined ionic liquid (D_{bulk} vs D_{confined}), interesting observations ought to be pointed out. Under confinement, the $Emim^+$ cations are accelerated and perform motions with considerably lower activation energy. Similar effects have been reported for confined ionic liquids and have been associated with the formation of coating layers on the pores' inner surfaces due to immobilization of chemical species from the fluid.[38−40] As the molecules involved in the formation of this layer do not exhibit strong interactions with the remaining mobile species as well as prevent the latter from interacting with the matrix, the remaining dynamic motions tend to be facilitated.[38−40] As previously mentioned, the differences between the confinement radii determined via our QENS analyses and the BET pore sizes determined in the previous works for the unloaded ^{11}BN do suggest the formation of a protective coating within the pores and are in line with the abovementioned hypothesis. Also, the extent of the differences between D_{bulk} and D_{confined} has been associated with the dimensions of the confinement geometry.[41] Here, however, even if the confinement radii are temperature-independent at temperatures above 290 K, the $D_{\mathrm{confined}}/D_{\mathrm{bulk}}$ ratio decreases as the temperature rises (see Figure 4a), suggesting that the thickness of the layer of immobile molecules near the pores' walls is not the sole contributor to the differences between D_{confined} and D_{bulk}.

In Figure 4b, the evolution of $D_{\mathrm{confined}}/D_{\mathrm{bulk}}$ as a function of the elastic fraction of $Emim^+$ ($A_0(Q)$ at Q = 1.1 Å^{-1}) is presented. At the temperature range where the confinement radii are temperature-independent (310−370 K), the evolution of $D_{\mathrm{confined}}/D_{\mathrm{bulk}}$ remarkably agrees with a simple exponential growth curve, as depicted by the dotted red line in the figure. In our case, for $\frac{D_{\mathrm{confined}}}{D_{\mathrm{bulk}}} = y_0 + A \times \exp(A_0(Q) - x_0)/t$, $y_0 = 1.14 \pm 0.02$, $A = 0.21 \pm 0.01$, and $t = 0.04 \pm 0.0007$ (note that while the data point collected at 290 K could also be included in a fit with the same equation, this would lead to exceptionally higher deviations in the determined parameters). Therefore, the possibility that $D_{\mathrm{confined}}/D_{\mathrm{bulk}}$ depends not only on the formation of the protective layer within the pores and the consequent changes in the confinement radius but also on the species involved in this coating cannot be disregarded.

■ CONCLUSIONS

Quasi-elastic neutron scattering (QENS) has been used to assess the long-range dynamics of the $Emim^+$ cations in solutions of LiTFSI in [Emim][TFSI] in bulk form and under confinement in an ^{11}BN matrix, where a quasi-liquid solid electrolyte (QLSE) was conceived.

For the bulk samples prepared with molalities of 0.5 and 2.5 mol·kg^{-1}, our results show how the large increase in LiTFSI concentration diminishes the mobility of the $Emim^+$ cations and increases the energy needed to activate their long-range motions. Namely, the 5-fold increase in LiTFSI concentration led to a 4-fold reduction in $Emim^+$'s diffusion coefficient.

The solution of LiTFSI in [Emim][TFSI] with a molality of 0.5 mol·kg^{-1} was then combined with ^{11}BN, leading to a QLSE, and the effects of the confinement in the $Emim^+$ dynamics have also been investigated. For this analysis, not only the diffusivities at different temperatures were assessed but also the fraction of the elastic scatters (that is, the molecules immobilized due to strong interactions with the matrix) and the confinement radii. Interestingly, above 290 K, the confinement radius is temperature-independent, regardless of the reduction in the elastic fraction in the QENS signal. Therefore, while Emim molecules engage in the formation of a protective coating in the pores' inner surfaces, they are likely replaced by other components of the fluid as the temperature increases. Of even greater importance, the diffusivities of the mobile fraction of the confined $Emim^+$ are highly increased in comparison with the corresponding bulk sample, and the activation energy for these motions is considerably reduced. We hypothesize that this effect is associated not only with the poor interactions between the mobile $Emim^+$ and the ^{11}BN matrix due to the layer formed by immobile species near the pores' surfaces but also with the changes in the content of Emim molecules that are involved in this process.

https://doi.org/10.1021/acs.jpcb.1c02383
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ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcb.1c02383.

Fitting of the QENS data for all samples including the dry ^{11}BN (PDF)

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Author Contributions

C.A.B. conceived the project and prepared the samples together with C.J.J., X.G.S., and M.P.P. with support from S.D. M.L.M. wrote the manuscript with input from all authors, performed the QENS experiments together with E.M. and with support from C.A.B., and analyzed the data together with E.M.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported as part of the Fluid Interface Reactions, Structures and Transport (FIRST) Center, an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences. This research used resources at the Spallation Neutron Source, a DOE Office of Science User Facility operated by the Oak Ridge National Laboratory. C.A.B., C.J.J., X.G.S., M.P.P., and S.D. were supported by the U.S. Department of Energy, Office of Science, Basic Energy Sciences, Materials Sciences and Engineering Division. This research used resources of the National Energy Research Scientific Computing Center, a DOE Office of Science User Facility supported by the Office of Science of the U.S. Department of Energy under contract no. DE-AC0205CH11231. Oak Ridge National Laboratory is managed by UT-Battelle, LLC, for U.S. DOE under contract no. DEAC05-00OR22725.

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